

MATH FORMULAS

Squares and Cubes and Square roots

1	1	1	1
2	4	8	1.414
3	9	27	1.732
4	16	64	2
5	25	125	2.236
6	36	216	2.449
7	49	343	2.646
8	64	512	2.828
9	81	729	3
10	100	1000	3.162
11	121	1331	3.317
12	144	1728	3.464
13	169	2197	3.606
14	196	2744	3.742
15	225	3375	3.873
16	256	4096	4
17	289	4913	4.123
18	324	5832	4.243
19	361	6859	4.359
20	400	8000	4.472
21	441	9261	
22	484	10648	
23	529	12167	
24	576	13824	
25	625	15625	
26	676	17576	
27	729	19683	
28	784	21952	
29	841	24389	
30	900	27000	
31	961	29791	
32	1024	32768	
33	1089	35937	
34	1156	39304	
35	1225	42875	
36	1296	46656	
37	1369	50653	
38	1444	54872	

FORMULAS

* product of LCM and HCF = product of two numbers

* Divisibility Rules

2 $\rightarrow$ even number (0, 2, 4, 6, 8)

3 $\rightarrow$ sum of all digits

4 $\rightarrow$ last two digits

5 $\rightarrow$ unit digit (0 & 5)

6 $\rightarrow$ divisible by 2 and 3.

7 $\rightarrow$ ABCD $\rightarrow$ $(ABC - 2D) \div 7 \checkmark$

8 $\rightarrow$ last three digits

9 $\rightarrow$ sum of all digits

10 $\rightarrow$ 2 & 5

11 $\rightarrow$ difference between sum of odd digits - sum of even digits

12 $\rightarrow$ 3 & 4

13 $\rightarrow$ abcd $\rightarrow$ $[abc + 4d] \div 13 \checkmark$

14 $\rightarrow$ 2 & 7

15 $\rightarrow$ 3 & 5

16 $\rightarrow$ last four digits

17 $\rightarrow$ ABCD $\rightarrow$ $[abc - 3d] \div 17 \checkmark$

18 $\rightarrow$ 2 & 9

19 $\rightarrow$ abcde $\rightarrow$ $[abcd + 2e] \div 19 \checkmark$

20 $\rightarrow$ 2 & 10

* prime numbers :-

2 3 5 7 11
13 17 19 23 29
31 37 41 43 47

53 59 61 67 71
73 79 83 89 97

< 50
15 $\checkmark$

> 50

10 $\checkmark$

1 to 100

25

101 103 107 109 113

127 131 137 149 151

157 163 167 173 179

181 191 193 197 199 211 223 227 229

* Sum of 'n' natural numbers = $\frac{n(n+1)}{2} = \sum n$

* Sum of squares of 'n' natural numbers = $\frac{n(n+1)(2n+1)}{6} = \sum n^2$

* $\sum n^3 = \frac{n^2(n+1)^2}{4} = (\sum n)^2$

* Sum of "Odd" n numbers = n^2
 "even" n numbers = $n^2 + n$

* LCM of a fraction = $\frac{\text{LCM of numerators}}{\text{HCF of denominators}}$

* HCF of a fraction = $\frac{\text{HCF of numerators}}{\text{LCM of denominators}}$

* $a^0 = 1, a^m \cdot a^n = a^{m+n}, (a^m)^n = a^{mn}, \frac{a^m}{a^n} = a^{m-n},$

$a^{-m} = \frac{1}{a^m}, (a \cdot b)^n = a^n \cdot b^n$

* Duplicate ratio of $a:b \Rightarrow a^2:b^2$

TriPLICATE " " " $\Rightarrow a^3:b^3$

Sub duplicate $\longrightarrow \sqrt{a}:\sqrt{b}$

Sub TriPLICATE $\longrightarrow \sqrt[3]{a}:\sqrt[3]{b}$

Reciprocate $\longrightarrow b:a$

* $\frac{a}{b}, \frac{c}{d}, \frac{e}{f} \rightarrow$ compound ratio $\Rightarrow ace = bdf$

* Componendo and Dividendo $\Rightarrow$ If $\frac{a}{b} = \frac{c}{d}$

then, $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

* A, B, C $\rightarrow$ invested $a:b:c$ for times $x:y:z$
 ratio of their profits $\Rightarrow ax:by:cz$

Algebra

$$\textcircled{1} \quad (a+b)^2 = a^2 + b^2 + 2ab$$

$$\textcircled{2} \quad (a-b)^2 = a^2 + b^2 - 2ab$$

$$\textcircled{3} \quad (a+b)^3 = a^3 + b^3 + 3ab(a+b)$$

$$\textcircled{4} \quad (a-b)^3 = a^3 - b^3 - 3ab(a-b)$$

$$\textcircled{5} \quad (a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$$

$$\textcircled{6} \quad (a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$\textcircled{7} \quad a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$\textcircled{8} \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$\begin{aligned} \textcircled{9} \quad a^3 + b^3 + c^3 - 3abc &= (a+b+c)[a^2 + b^2 + c^2 - ab - bc - ca] \\ &= \frac{1}{2} [(a-b)^2 + (b-c)^2 + (c-a)^2] \end{aligned}$$

If $a+b+c=0$, then $a^3 + b^3 + c^3 = 3abc$

If $a=b=c$ then $a^3 + b^3 + c^3 = 3abc$ and
 $a^2 + b^2 + c^2 - ab - bc - ca = 0$

$$\textcircled{10} \quad x + \frac{1}{x} = a \rightarrow x^2 + \frac{1}{x^2} = a^2 - 2$$

$$x^3 + \frac{1}{x^3} = a^3 - 3a$$

$$x^4 + \frac{1}{x^4} = a^4 - 4a^2 + 2$$

$$x^5 + \frac{1}{x^5} = a^5 - 5a^3 + 5a$$

$$x^6 + \frac{1}{x^6} = a^6 - 6a^4 + 9a^2 - 2$$

$$\textcircled{11} \quad x - \frac{1}{x} = a \rightarrow x^2 + \frac{1}{x^2} = a^2 + 2$$

$$x^3 - \frac{1}{x^3} = a^3 + 3a$$

$$x^4 + \frac{1}{x^4} = a^4 + 4a^2 + 2$$

$$x^5 - \frac{1}{x^5} = a^5 + 5a^3 + 5a$$

$$x^6 + \frac{1}{x^6} = a^6 + 6a^4 + 9a^2 + 2$$

$$(12) \quad x + 1/x = \sqrt{3} \Rightarrow x^3 + 1/x^3 = 0 ; x^6 = -1$$

$$(13) \quad a^4 + a^2 + 1 = (a^2 + a + 1)(a^2 - a + 1)$$

$$a^4 + a^2 b^2 + b^4 = (a^2 + ab + b^2)(a^2 - ab + b^2)$$

$$(14) \quad x + 1/x = k \Rightarrow x - 1/x = \sqrt{k^2 - 4}$$

$$x + 1/x = 1 \Rightarrow x^2 - x + 1 = 0 ; x^3 + 1 = 0$$

$$x + 1/x = -1 \Rightarrow x^2 + x + 1 = 0 ; x^3 - 1 = 0$$

$$(15) \quad (a+b+c)^3 = a^3 + b^3 + c^3 + 3(a+b)(b+c)(c+a)$$

$$(16) \quad (a+b)^4 = a^4 + b^4 + 4a^3b + 6a^2b^2 + 4ab^3$$

$$(a-b)^4 = a^4 + b^4 - 4a^3b + 6a^2b^2 - 4ab^3$$

$$a^4 - b^4 = (a-b)(a+b)(a^2 + b^2)$$

$$a^5 - b^5 = (a-b)(a^4 + a^3b + a^2b^2 + ab^3 + b^4)$$

$$(17) \quad a^2 + b^2 = x, \quad a^2 - b^2 = y$$

$$\text{then, } ab = \frac{x-y}{2}$$

Quadratic $\rightarrow$ roots

$$ax^2 + bx + c \rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\begin{aligned} \alpha + \beta &= -b/a \\ \alpha\beta &= c/a \end{aligned}$$

$$* \quad ax^3 + bx^2 + cx + d$$

$\downarrow$

$$\alpha + \beta + \gamma = -b/a$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = c/a$$

$$\alpha\beta\gamma = -d/a$$

SI & CI

$$\star SI = \frac{P \cdot R \cdot T}{100}$$

$$CI = P \left(1 + \frac{R}{100} \right)^2$$

$$A = P \left(1 + \frac{RT}{100} \right)$$

$$\longrightarrow CI = P \left(1 + \frac{r}{200} \right)^{2n} \rightarrow \text{Half yearly}$$

$$\longrightarrow CI = P \left(1 + \frac{r}{400} \right)^{4n} \rightarrow \text{Quarterly}$$

$$\longrightarrow CI = P \left(1 + \frac{r}{1200} \right)^{12n} \rightarrow \text{monthly.}$$

$$\text{If } T = 4\frac{3}{5} \text{ years}$$

$$\hookrightarrow CI = P \left(1 + \frac{r}{100} \right)^4 \cdot \left(1 + \frac{3r}{500} \right)$$

$\star$ Relation between CI and SI for 2 years

$$\frac{CI}{SI} = \frac{200 + r}{200}$$

$\star$ Difference between CI and SI

$$\text{for 2 years} \rightarrow P \left(\frac{r}{100} \right)^2$$

$$\text{for 3 years} \rightarrow \frac{P r^2 (300 + r)}{100^3}$$

$$\rightarrow \frac{P r^2}{100^2} \left[\frac{300 + r}{100} \right]$$

* Avg of 'x' numbers = a
 Avg of 'y' numbers = b } avg of x+y = $\frac{2a+2b}{2+y}$

* $\frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2}$

* A does work in x days

B $\longrightarrow$ y

(A+B) $\rightarrow$ can complete in $\frac{xy}{x+y}$ days.

* A, B, C $\rightarrow$ x, y, z

(A+B+C) $\Rightarrow$ $\frac{xyz}{xy+yz+zx}$ days

* $\left. \begin{array}{l} (A+B) \rightarrow x \text{ days} \\ (B+C) \rightarrow y \text{ days} \\ (C+A) \rightarrow z \text{ days} \end{array} \right\} \Rightarrow (A+B+C) = \frac{2xyz}{xy+yz+zx} \text{ days}$

* Speed = $\frac{\text{Distance}}{\text{Time}}$

T $\rightarrow$ 1st S₁, 2nd S₂

D = $\frac{2TS_1S_2}{S_1+S_2}$

Avg speed of x, y = $\frac{2xy}{x+y}$ km/hr

* Boat stream

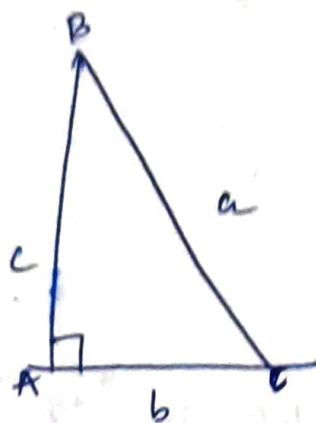
① Downstream (v) = x+y

② Up stream (u) = x-y

③ Speed of boat still in water = $\frac{1}{2}(u+v) = x$
 [Boat $\leftrightarrow$ x]

④ Speed of stream = y = $\frac{1}{2}(v-u)$
 [stream $\leftrightarrow$ y]

Pythagorean Triplets



$$\Rightarrow a^2 = b^2 + c^2$$

* • (3, 4, 5)	• (5, 12, 13)	• (21, 72, 75)	• (1, $\sqrt{3}$, 2)
• (6, 8, 10)	• (10, 24, 26)	• (8, 15, 17)	• (2n, n^2-1 , n^2+1)
• (9, 12, 15)	• (7, 24, 25)	• (9, 40, 41)	• ($\frac{n^2+1}{2}$, $\frac{n^2-1}{2}$, n)
• (12, 16, 20)	• (14, 48, 50)	• (11, 60, 61)	• (2ab, a^2-b^2 , a^2+b^2)
• (15, 20, 25)	• (3.5, 12, 12.5)	• (1, 1, $\sqrt{2}$)	• (12, 35, 37)

Coordinate Geometry

* Distance = $\sqrt{(y_2 - y_1)^2 + (x_2 - x_1)^2}$

$$ax + by + c = 0$$



from origin $d = \left| \frac{c}{\sqrt{a^2 + b^2}} \right|$

* $x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}$, $y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$

midpoints $\rightarrow x = \frac{x_1 + x_2}{2}$, $y = \frac{y_1 + y_2}{2}$

* $m = \text{slope} = \frac{y_2 - y_1}{x_2 - x_1}$

$$\tan \theta = \frac{m_2 - m_1}{1 + m_1 m_2}$$

$$y = mx + c$$

* Unique solution

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

* Infinite solution

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

* No solution

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

* $m_1 = m_2$ (parallel)

* $m_1 m_2 = -1$ ($\perp$ lines)

Mensuration - 2D

- * ① Triangle $a+b>c$; sum of angles = 180°
 $a-b<c$
perimeter $>$ sum of its median

$$\text{Area} = \frac{1}{2}bh = \sqrt{s(s-a)(s-b)(s-c)} \Rightarrow s = \frac{a+b+c}{2}$$

$$= \frac{1}{2}bc \sin A$$

$$= r s \quad (r = \text{in radius}, s = \text{semi perimeter})$$

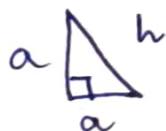
$$= \frac{abc}{4R} \quad (R = \text{circum radius})$$

② Isosceles Δ^e $\text{Area} = \frac{b\sqrt{4a^2-b^2}}{4}$

$$\text{Height} = \frac{\sqrt{4a^2-b^2}}{2}$$

$$\text{Perimeter} = 2a+b$$

- ③ Isosceles right angled Δ^e

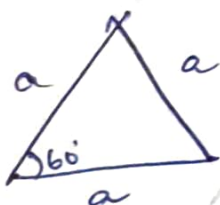


$$h = a\sqrt{2}$$

$$A = \frac{1}{2}a^2$$

$$P = 2a+h = 2s$$

- ④ Equilateral triangle



$$\theta = 60^\circ$$

$$\text{Area} = \frac{\sqrt{3}}{4}a^2$$

$$\text{Median} = \frac{\sqrt{3}}{2}a$$

$$\text{Perimeter} = 3a$$

$$\text{Height} = \frac{\sqrt{3}}{2}a$$

$$\text{In radius} = \frac{a}{2\sqrt{3}}$$

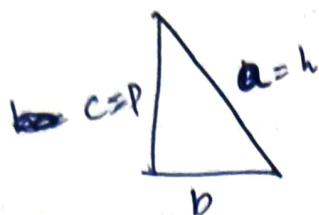
$$\text{Circumradius} = \frac{a}{\sqrt{3}}$$

⑤ Right angled Δ^c

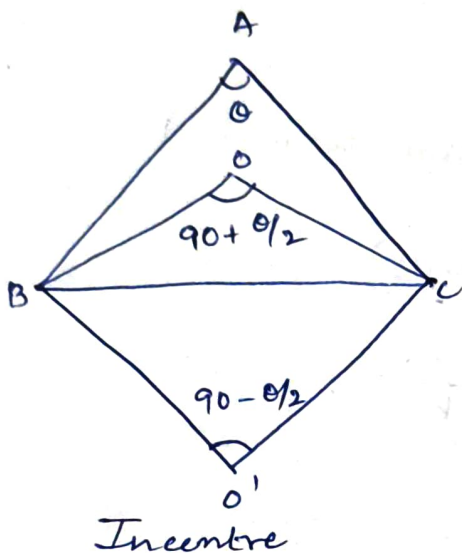
Inradius $\Rightarrow r = \frac{p+b-h}{2}$

$A = \frac{1}{2}bh$

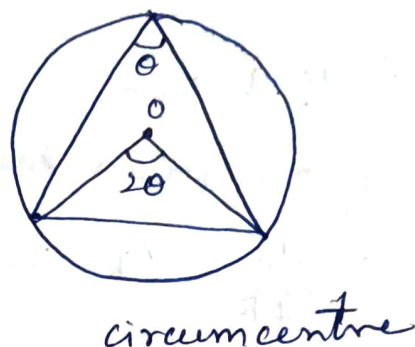
$a^2 = b^2 + c^2$



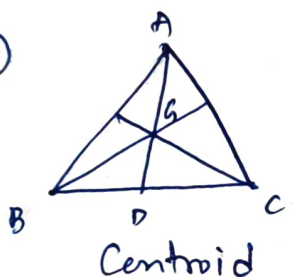
⑥



⑦

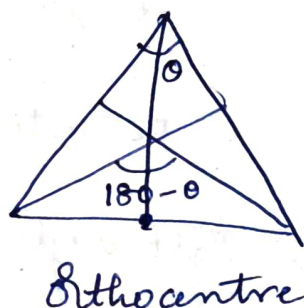


⑧



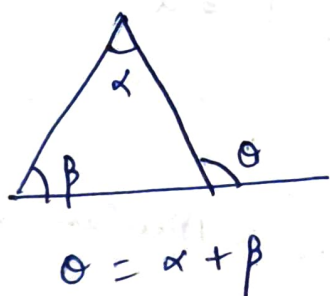
$\frac{AG}{GD} = \frac{2}{1}$

⑨

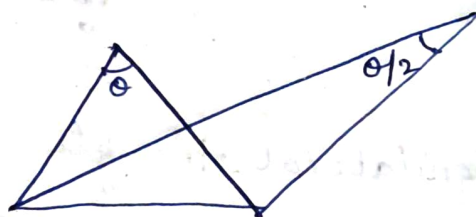


Orthocentre

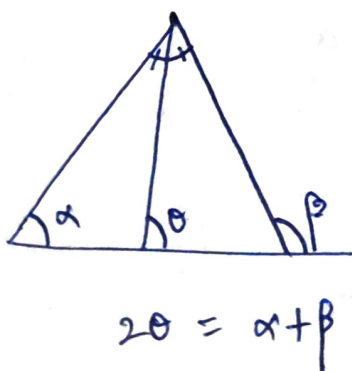
⑩



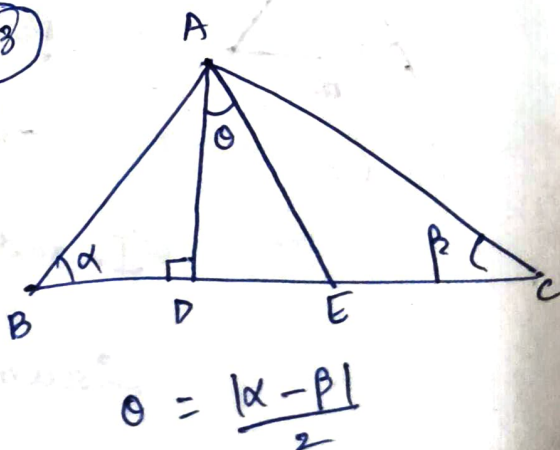
⑪



⑫



⑬

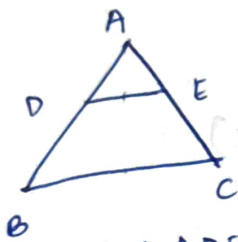


$$(14) \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

(15) Thales Theorem

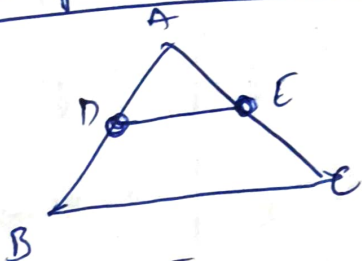


$$\frac{AD}{DB} = \frac{AE}{EC} = \frac{DE}{BC}$$

$$\triangle ADE \sim \triangle ABC$$

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle ABC)} = \left(\frac{AD}{DB}\right)^2 = \left(\frac{AE}{EC}\right)^2 = \left(\frac{DE}{BC}\right)^2$$

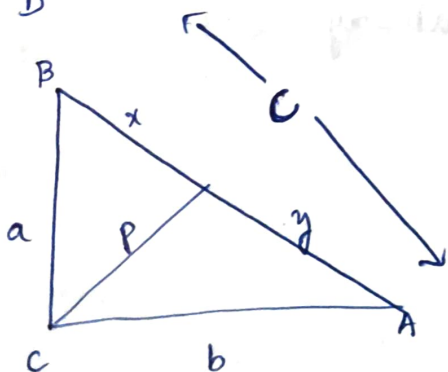
(16) Midpoint Theorem



D, E $\rightarrow$ midpoints of AB, AC

$$DE = \frac{BC}{2}$$

(17)



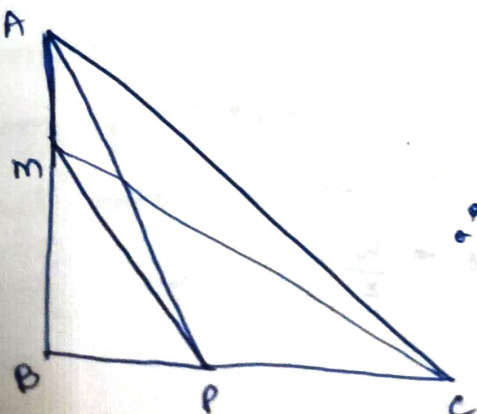
$$\Rightarrow \begin{aligned} (1) \quad a^2 &= x \cdot c \\ b^2 &= y \cdot c \end{aligned}$$

$$(2) \quad p^2 = x \cdot y$$

$$(3) \quad a \times b = p \times c$$

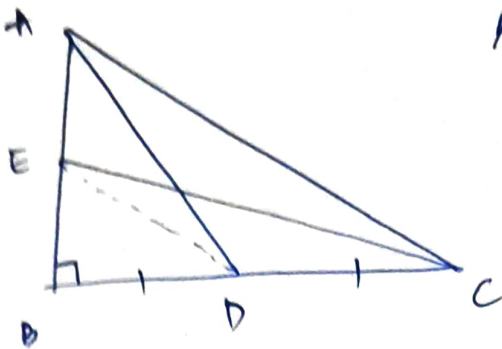
$$(4) \quad \frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$$

(18)



$$\therefore AP^2 + MC^2 = AC^2 + MP^2$$

(19.)

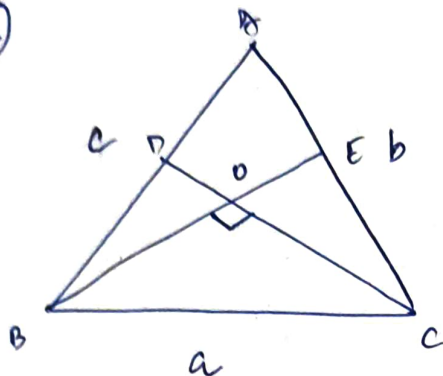


$AD, CE \rightarrow$ medians

then,

- $4(AD^2 + CE^2) = 5AC^2$
- $AD^2 + CE^2 = 5ED^2$
- $AD^2 + CE^2 = AC^2 + ED^2$

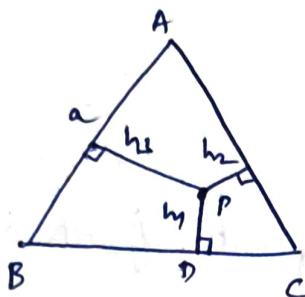
(20.)



$$b^2 + c^2 = 5a^2$$

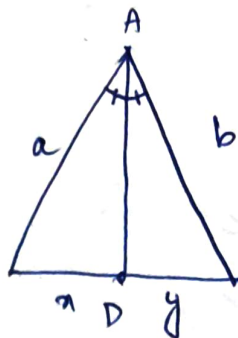
(medians $\rightarrow BE, DC$)

(21)



$$h_1 + h_2 + h_3 = \frac{\sqrt{3}}{2} a$$

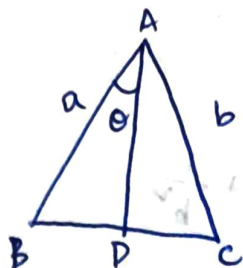
(22)



angular bisector

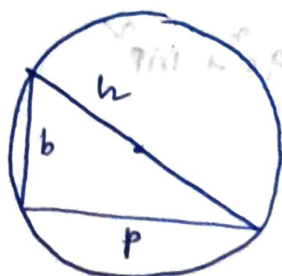
$$AD^2 = ab - xy$$

(23)



$$AD = \frac{2ab \cos \theta}{a+b}$$

(24.)



$$\text{circumradius} = \frac{h}{2}$$

(25) Distance between circumcentre and incentre = $\sqrt{R^2 - 2r^2}$

(26) Apollonius Theorem



$$AB^2 + AC^2 = 2(AD^2 + CD^2)$$

length of median,

$$AD = \frac{1}{2} \sqrt{2AC^2 + 2AB^2 - BC^2}$$

$$\frac{AG^2 + BG^2 + CG^2}{AB^2 + BC^2 + CA^2} = \frac{1}{3}$$

(27)

(3D)

Figure	Volume	TSA	CSA
Cube	a^3	$6a^2$	$4a^2$
Cuboid	lbh	$2(lb + bh + hl)$	$2(lh + bh)$
Cylinder	$\pi r^2 h$	$2\pi r(r + h)$	$2\pi rh$
Cone	$\frac{1}{3} \pi r^2 h$	$\pi r(r + l)$	πrl
Sphere	$\frac{4}{3} \pi r^3$	$4\pi r^2$	$4\pi r^2$
Hemisphere	$\frac{2}{3} \pi r^3$	$3\pi r^2$	$2\pi r^2$
Frustum of cone	$\frac{1}{3} \pi [R^2 + r^2 + Rr]h$	$\pi (R + r)l + \pi (R^2 + r^2)$	$\pi (R + r)l$
Hollow sphere	$\frac{4}{3} \pi (R^3 - r^3)$	$4\pi (R^2 - r^2)$	$4\pi (R^2 - r^2)$
Prism	Base area $\times h$	LSA + 2(Base area)	Base peri $\times h$
Pyramid	$\frac{1}{3}$ Base area $\times h$	LSA + Base area	$\frac{1}{2} B \cdot P \times l$
Hexagon			

$$\hookrightarrow \frac{3\sqrt{3}}{2} a^2 = A, \quad l = \sqrt{h^2 + r^2}$$

Octagon

$$A = 2(\sqrt{2} + 1)a^2$$

* sum of all internal angles of polygon of n sides = $(n-2) \cdot 180^\circ$

* sum of all exterior angles of polygon = 360°

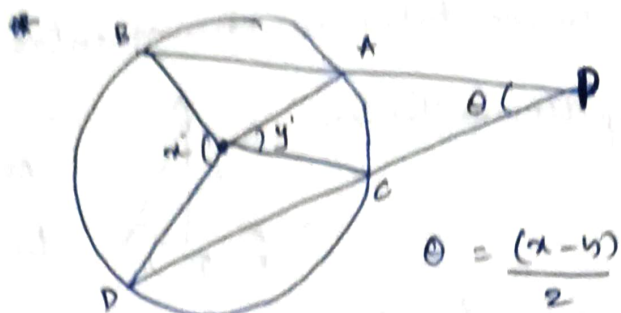
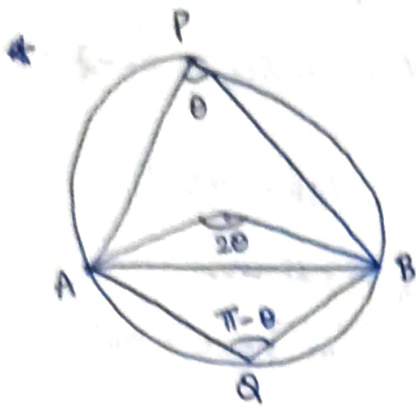
* each interior angle = $\frac{(n-2)180^\circ}{n}$; each exterior angle = $\frac{360^\circ}{n}$

* no. of diagonals = $\frac{n(n-3)}{2}$

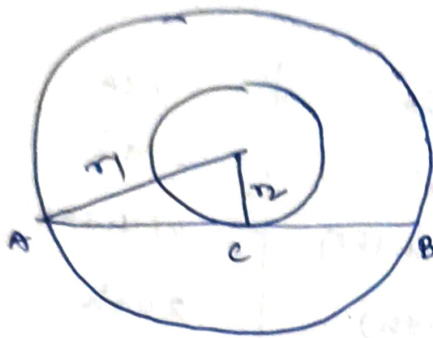
$$* \text{area} = \frac{n \cdot a^2}{4} \cot \frac{180^\circ}{n}$$

* sum of outer angle = $180(n-4)$

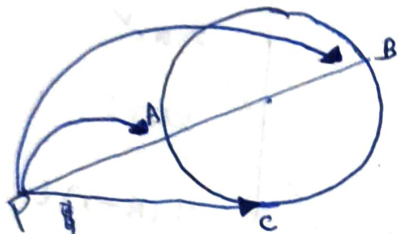
$$* \frac{\text{interior}}{\text{exterior}} = \frac{(n/2 - 1)}{1}$$



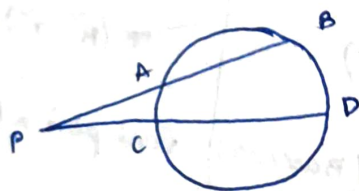
$$\theta = \frac{(x - y)}{2}$$



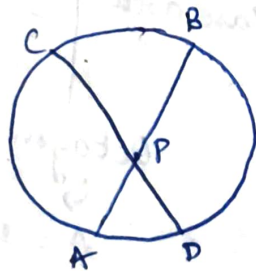
$$AB = 2AC = 2\sqrt{r_1^2 - r_2^2}$$



$$PA \cdot PB = PC^2$$



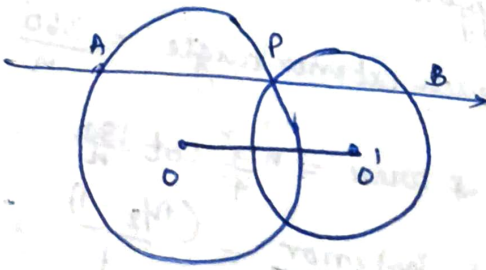
$$PA \times PB = PC \times PD$$



$$PA \times PB = PC \times PD$$

$$r = \frac{\sqrt{AP^2 + PD^2 + CP^2 + BP^2}}{2}$$

$$\text{If } \angle APD = 90^\circ, r = \frac{\sqrt{BC^2 + AD^2}}{2}$$



$$AB = 2 \times OO'$$

* length of Direct common tangent

$$L = \sqrt{d^2 - (r_2 - r_1)^2}$$



* length of Transverse common tangent

$$L = \sqrt{d^2 - (R_1 + R_2)^2}$$



* $\rightarrow DCT = 2\sqrt{r_1 r_2}$



$$A = \pi r^2$$

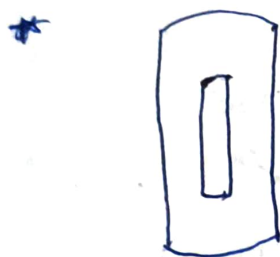
$$d = 2r$$

$$P = 2\pi r$$



$$A = \frac{\pi r^2}{2}$$

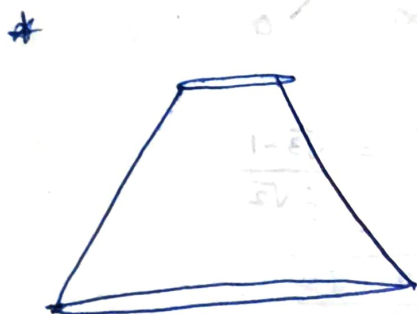
$$P = \pi r + 2r$$



$$V = \pi h (R^2 - r^2)$$

$$TSA = \pi (R^2 - r^2) h$$

$$CSA = 2\pi h (r + R)$$



$$V = \frac{1}{3} \pi [R^2 + Rr + r^2] h$$

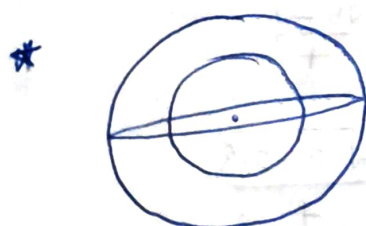
$$L^2 = (R - r)^2 + h^2$$

$$LSA = \pi (R + r) L$$

$$TSA = \pi (R^2 + r^2 + RL + rL)$$

$$\frac{r}{R} = \frac{H-h}{H}$$

remaining cut.

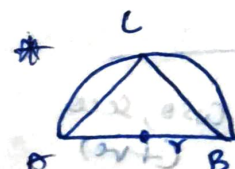


$$TSA = 4\pi (R^2 + r^2)$$

$$V = \frac{4}{3} \pi (R^3 - r^3)$$

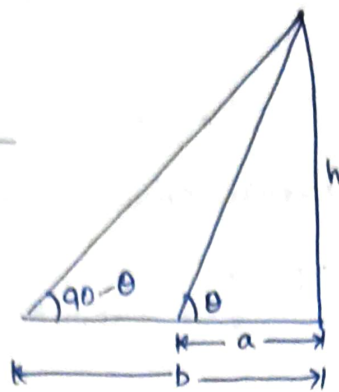
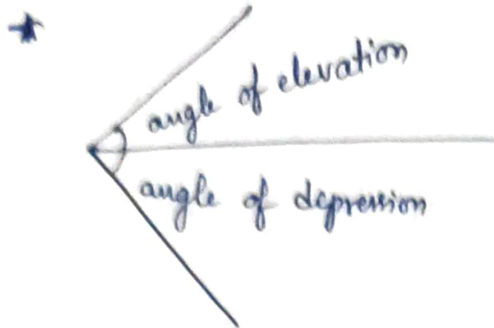
* ring $\rightarrow A = \pi (R^2 - r^2)$

$$P = 2\pi (R + r)$$

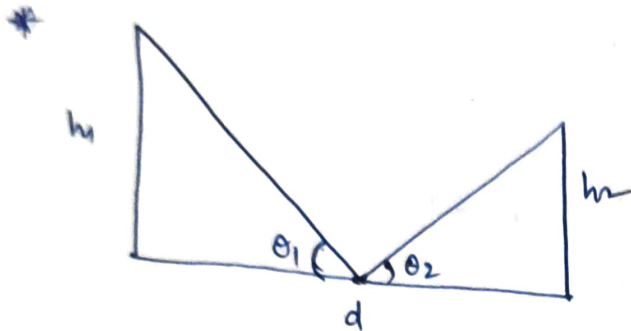


Area of $\Delta = r^2$

* $n = \frac{d}{2\pi r}$ (no. of revolutions made by wheel)



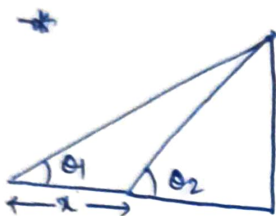
$$h = \sqrt{ab}$$



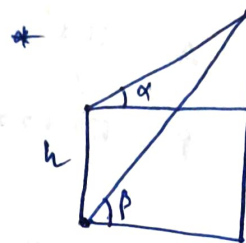
use $\tan \theta$.

If $h_1 = h_2$

$$h = \frac{d}{\cot \theta_1 + \cot \theta_2}$$

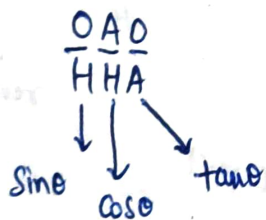


$$h = \frac{x}{\cot \theta_1 - \cot \theta_2}$$



$$H = \frac{h \tan \beta}{\tan \beta - \tan \alpha}$$

	0°	30°	45°	60°	90°	180°	270°	360°
sin θ	0	1/2	1/√2	√3/2	1	0	-1	0
cos θ	1	√3/2	1/√2	1/2	0	-1	0	1
tan θ	0	1/√3	1	√3	∞	0	∞	0



★ $\sin 15^\circ = \cos 75^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}}$

★ $\sin 22.5^\circ = \frac{\sqrt{2-\sqrt{2}}}{2}$

★ $\sin 18^\circ = \cos 72^\circ = \frac{\sqrt{5}-1}{4}$

★ $\sin 36^\circ = \cos 54^\circ = \frac{\sqrt{10-2\sqrt{5}}}{4}$

sin θ, cosec θ (+ve)	(+) all
tan θ, cot θ (+ve)	cos θ, sec θ (+ve)

$$* A + B = 90^\circ$$

$$\sin A = \cos B, \tan A \tan B = 1, \cot A \cdot \cot B = 1$$

$$* \begin{aligned} \sin(A+B) &= \sin A \cos B + \cos A \sin B \\ \sin(A-B) &= \sin A \cos B - \cos A \sin B \\ \cos(A+B) &= \cos A \cos B - \sin A \sin B \\ \cos(A-B) &= \cos A \cos B + \sin A \sin B \end{aligned}$$

$$\begin{aligned} \sin^2 \theta + \cos^2 \theta &= 1 \\ \sec^2 \theta - \tan^2 \theta &= 1 \\ \operatorname{cosec}^2 \theta - \cot^2 \theta &= 1 \end{aligned}$$

$$* \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}, \quad \tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$* \tan 2A = \frac{2 \tan A}{1 - \tan^2 A}, \quad \tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$$

$$* \sin 3A = 3 \sin A - 4 \sin^3 A, \quad \cos 3A = 4 \cos^3 A - 3 \cos A$$

$$* \sin C + \sin D = 2 \sin \left(\frac{C+D}{2} \right) \cos \left(\frac{C-D}{2} \right)$$

$$* \sin C - \sin D = 2 \cos \left(\frac{C+D}{2} \right) \sin \left(\frac{C-D}{2} \right)$$

$$* \cos C + \cos D = 2 \cos \left(\frac{C+D}{2} \right) \cos \left(\frac{C-D}{2} \right)$$

$$* \cos C - \cos D = 2 \sin \left(\frac{C+D}{2} \right) \sin \left(\frac{C-D}{2} \right)$$

$$* \sin \theta \cdot \sin 2\theta \cdot \sin 4\theta = \frac{1}{4} \sin 3\theta$$

$$* \cos \theta \cdot \cos 2\theta \cdot \cos 4\theta = \frac{1}{4} \cos 3\theta$$

$$* \tan \theta \cdot \tan 2\theta \cdot \tan 4\theta = \tan 3\theta$$

$$* \sin 2\theta = 2 \sin \theta \cos \theta$$

$$* \cos 2\theta = \cos^2 \theta - \sin^2 \theta = 1 - 2 \sin^2 \theta = 2 \cos^2 \theta - 1 = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$$

$$* \tan \frac{\theta}{2} = \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$$

$$* \cot \frac{\theta}{2} = \frac{1 + \cos \theta}{\sin \theta}$$

$$* \sec \theta + \tan \theta = \sqrt{\frac{1 + \sin \theta}{1 - \sin \theta}}$$

$$* \sec^4 \theta - \sec^2 \theta = \tan^2 \theta + \tan^4 \theta \quad * 1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}$$

$$* 1 - \cos \theta = 2 \sin^2 \frac{\theta}{2} \quad ; \quad \tan \frac{\theta}{2} = \frac{1 - \cos \theta}{\sin \theta}$$

$$\star \cot \alpha = \tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + 8 \tan 8\alpha$$

$$\star 3 \cot 3\theta = \cot \theta + \cot (60+\theta) + \cot (120+\theta)$$

$$\star (1 + \cot \theta - \operatorname{cosec} \theta) (1 + \tan \theta \sec \theta) = 2$$

Statistics

$$\star \text{Mean} = \frac{\text{sum of observations}}{\text{No. of observations}} = \frac{\sum x_i}{n}$$

$$\star \text{A.M} = \frac{a+b}{2} \quad \text{G.P} \rightarrow \text{A.M} = \frac{a(r^n - 1)}{(r - 1)}$$

$$\text{G.M} = \sqrt{ab}$$

$$\text{H.M} = \frac{2ab}{a+b}$$

$$\star \text{Median} = \text{middle value}$$

or

$$= \left(\frac{n}{2} + 1\right)^{\text{th}} \text{observation}$$

$$\star \text{Mode} = \text{most repeated value.}$$

$$\star \text{Empirical relation:- } \boxed{\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}}$$

$$\star \text{Range} = \text{Difference [Highest - lowest]}$$

$$\star \text{Mean deviation} = \frac{1}{n} \sum_{i=1}^n |x_i - a|$$

$$\star \text{Variance} = \sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{\sum x_i^2}{n} - \left(\frac{1}{n} \sum x_i\right)^2$$

$$\star \text{S.D} = \sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\star \text{Median} = l + \left(\frac{N/2 - c}{f}\right) i$$

$$\star \text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) i$$

$l \rightarrow$ lower limit

$N \rightarrow$ sum of frequencies

$i \rightarrow$ width of class

$c \rightarrow$ cumulative frequency

$f_0 \rightarrow$ modal class - 1

$f_1 \rightarrow$ modal class

* Permutation, Combination and Probability

* $n! = n(n-1)(n-2)(n-3) \dots 3 \cdot 2 \cdot 1 = 1 \cdot 2 \cdot 3 \dots n$

* $(m+n)! \neq m! + n!$

$(m-n)! \neq m! - n!$

$(mn)! \neq (m!)(n!)$

$\left(\frac{m}{n}\right)! \neq \frac{m!}{n!}$

* $0! = 1$

$6! = 720$

* ${}^nC_r = {}^nC_{n-r}$

$1! = 1$

$7! = 5040$

* permutations - repetition

$2! = 2$

$8! = 40320$

↓
 n^r

$3! = 6$

$9! = 362880$

$4! = 24$

$10! = 3628800$

* $0 \leq P \leq 1$

$5! = 120$

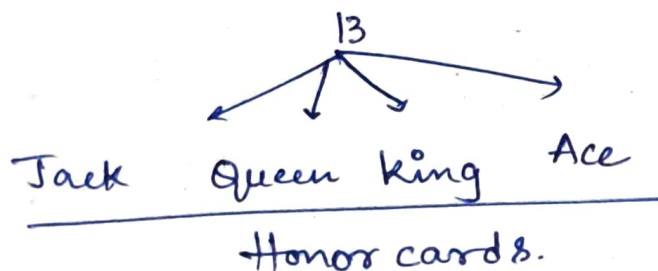
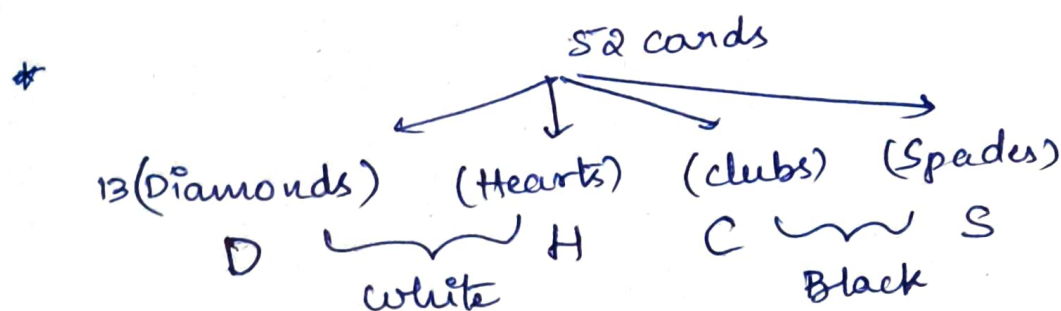
* ${}^nP_r = \frac{n!}{(n-r)!}$

* ${}^nC_r = \frac{n!}{r!(n-r)!}$

${}^nP_n = n!$

${}^nC_0 = 1 = {}^nC_n = {}^nC_1$

* ${}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n - 1$

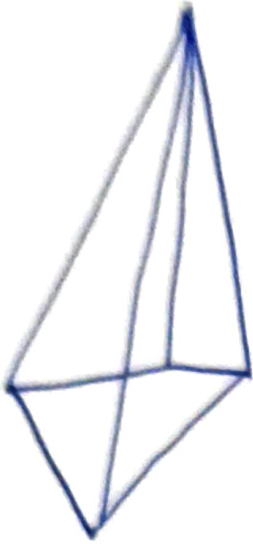


2
3
4
5
6
7
8
9
10

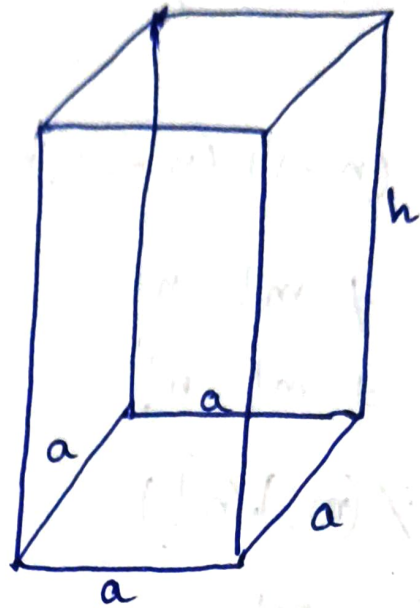
9 + 4

* $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

* $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$



Square pyramid



Square prism

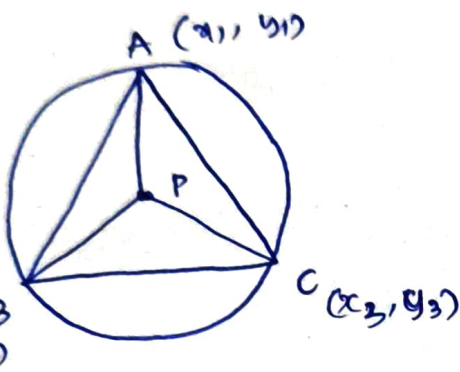
$$\begin{aligned}\text{Volume} &= \text{Area of Base} \times \text{Height} \\ &= a^2 \times h\end{aligned}$$

$$\begin{aligned}\text{CSA} &= \text{perimeter of Base} \times \text{Height} \\ &= 4ah\end{aligned}$$

$$\begin{aligned}\text{TSA} &= \text{CSA} + 2 \cdot (\text{Base Area}) \\ &= 4ah + 2a^2\end{aligned}$$

* Coefficient of Variation = $\frac{\text{Standard Deviation}}{\text{Arithmetic Mean}} \times 100$

Coefficient of skewness = $\frac{(\text{Arithmetic Mean} - \text{Mode})}{\text{S.D}}$



coordinates of circumcenter = ?

$P \equiv (a, b)$

$(x_2 - a)^2 + (y_2 - b)^2 = (a - x_3)^2 + (b - y_3)^2$

* The hour hand moves 0.5° per minute.

* Angle between hour hand and minute hand

$\theta = \left| \frac{11}{2} m - 30H \right|$

* A wire is bent to form a square of area $A \text{ cm}^2$, If the same wire is bent to form a ring, then area is enclosed by that ring = $\frac{4A}{\pi}$